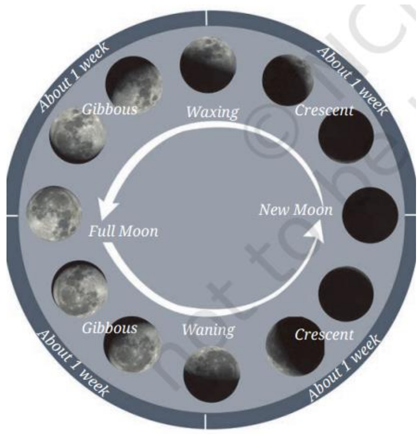
Chapter – 11: Keeping Time with the Skies

- Makar Sankranti – Meera – Ahmedabad – *Patang Mahotsav* – International Kite Festival
- Looked up – sky – colourful kites – notice moon – daytime – surprised – always thought – moon – nighttime
- Also – moon – not always – full circle – not much surprise
- Knew – shape change – every night – remember learning – moon – spherical – reflect sunlight – shine
- Whole moon – not visible – every night – wonders – reason – lunar eclipse – BUT – eclipses – rare, brief

How Does the Moon's Appearance Change and Why?

- **Watch** moon – **understand** – changing appearance – over month
- May have done – similar act. – earlier – BUT – this time – more detail
- Begin act. – sunrise – full moon day – easiest – spot moon – sky
- Activity –
  - **Spot** – moon – sunrise – western dir. – 1^{st} day after full moon
  - Make table – document following –
    - Date
    - Time – sunrise, sunset
    - Shade – corresponding circle – show – bright portion – moon
  - 2^{nd} day onward – also document –
    - Size – bright portion – increasing, decreasing?
    - Moon – farther, closer – sun – previous day
  - 15 days later – moon -not visible – sunrise, sunset – next 15 days – carry – this act. – sunset
- **Analyse** – recorded data – moon – size, position, visibility

Phases of the Moon



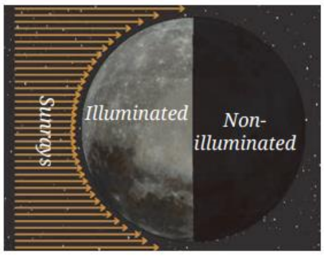
- Obs. – bright portion – moon – decrease – full to half – 1 week
- Bright portion – shrink – another week – until – no visibility
- This 2-week period – **waning** period – moon
- Diff. names – moon's visible shapes – this cycle
- Day – moon – full bright circle – **full moon** day (*Purnima*) – day – not visible – **new moon** day (*Amavasya*
- After new moon – bright side – grow – half circle – 1 week – full circle – another week
- Period – bright part – moon – increase – **waxing** period
- India – waning period – *Krishna Paksha* – BUT – waxing period – *Shukla Paksha*
- Changing shapes – bright portion – moon – **phases of moon**

Locating the Moon

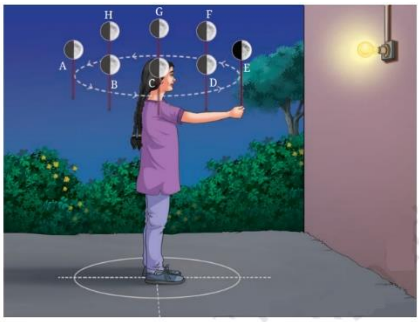
- Check moon – same time – successive days – visible – diff. part
- Full moon day – moon – opp. sun – sun – rise – east – moon – set – west
- Subsequent mornings – sunrise – bright part – decrease – moon – moves closer – sun
- Bright part – moon – half circle – moon – overhead – sunrise
- Few days later – crescent moon – even closer – sun

- Knowing – phase, waxing, waning – help us – spot moon – any day
- Waxing moon – easiest spot – sunrise – waning moon – sunrise
- These shifts – reason – moon – rise, set – diff. time – sun
- **A Step Further**
    - Many people – believe – sun set – then moon rise – BUT – not always true
    - Look – local newspaper – Positional Astronomy Centre website – moonrise time – your area
    - Check times – several days – obs. – moonrise – 50 min. later – every day
    - Sometimes – moonrise – afternoon – 2:00 – 4:00 PM – spot moon – eastern sky – daylight
    - Visit – 30 min. later – listed time – moon – climb high – visible

Making sense of our observations

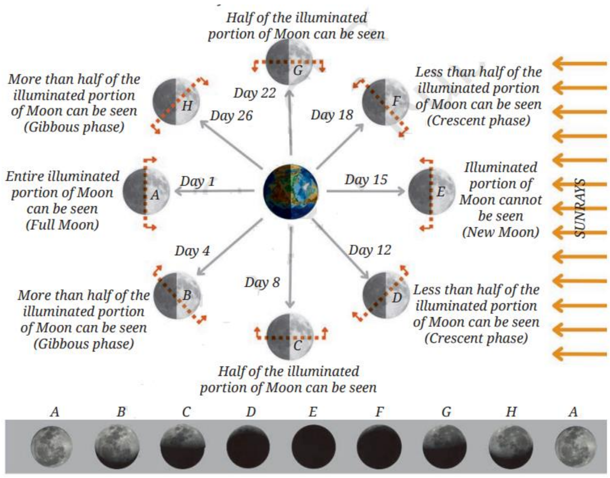


- Shape – moon – no change – only visibility – change
- Recall learning – earlier – moon – no emission – own light – BUT – shine – reflected sunlight
- Half – moon – face sun – receive sunlight – illuminated – other half – no sunlight – no illumination
- Moon – revolve around earth – 1 half – always face earth
- Portion – facing earth – not always illuminated – illuminated portion – always visible
- Sometimes – entire illuminated portion – face earth – other times – only part
- Such times – illuminated portion – not full circle
- New moon day – no illuminated portion – only non-illuminated portion – face earth
- This reason – moon – diff. appearance – diff. days
- Perform act. – understand – illuminated portion – moon – changes – position changes w.r.t. sun
- Activity –
    - Take – small soft ball – insert stick – represent moon
    - Go – dark open place – ask – teacher, guardian – shine torchlight – 3m away – represent sunlight – your head – represent earth
    - Hold ball – arm's length – 1 hand – slightly above head – keep ball – position E – dir. – lamp – portion – ball – illuminated
    - Turn around – anti-clockwise – arms stretched outwards – keep looking – ball – shape – illuminated portion – change – separating line – curved?
    - Your obs. – similar? – changing shape – illuminated portion – shape – illuminated portion – ball – as seen – change – depend on – position – ball – w.r.t. lamp

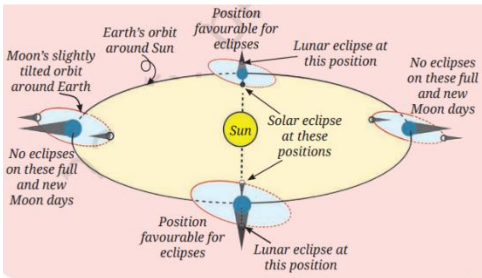


- Ball – held opp. – lamp dir. (A) – you face – entire illuminated portion – full moon day
- Other hand – ball – help towards – lamp dir. (E) – you face – non-illuminated portion – new moon day
- Notice – separating line – illuminated, non-illuminated portions – curved – similar – illuminated shape – moon – visible – earth
- Use obs. – above act. – understand phases – moon
- Fig. – moon pos. – correspond – diff. pos. – ball – also shown – earth, sunrays
- Shown – fig. – moon – 1 revolution – earth – A to H – back to A – 1 month – side – moon – face sun – illuminated
- Portion – moon – face earth – marked – orange dash line, arrow
- Illuminated portion – this part – visible – earth
- Pos. B, H – 50% + portion – visible – **gibbous phase**

- Pos. D, F – 50% - portion – visible – **crescent phase**
- Change – fraction – visible portion – moon – visible – earth – reason – moon phases
- Phases – visible – earth – diff. pos. – moon
- A to C to E – waning phase – E to G to A – waxing phase
- Rotation period – earth – 1 day – much smaller – compare – revolution period – moon – 1 month
- Given day – people – diff. parts – earth – same phase moon
- Fig. – new moon day – moon – closest – sun – appears farthest – full moon day
- 1st act. – also obs. – moon pos. – sunrise – shift – successive days
- Reason shown – fig. – moon – move fwd. – orbit – earth rotate – axis – 24 hrs.
- Earth – rotate more – moon visible – same spot – sky

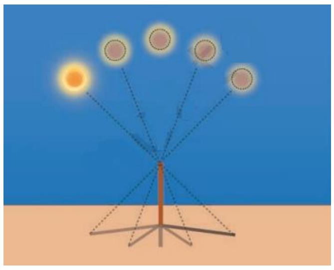


- **A Step Further**
  - Moon phase – not because – earth’s shadow
  - Learnt – moon phases – reason – rel. change – orientation – sun, moon, earth – moon revolve – earth
  - Earth’s shadow – moon – cause – lunar eclipse – not moon phase – learnt earlier
  - Lunar eclipse – only happen – full moon day – solar eclipse – only happen – new moon day
  - BUT – not every month – reason – tilt – moon’s orbit – w.r.t. earth’s orbit



How Did Calendars Come Into Existence?

- Learnt earlier – viewed – earth – sun – rise – east dir. – set – west dir.
- Periodic motion – sun – primary reason – rotation – earth – own axis
- Natural cycle – sun – reason – earth rotation – foundation – day – unit – measure time
- Avg. time – sun – highest pos. – 1 day – highest position – next day – 24 hrs. – **mean solar day**
- Highest pos. – sun – sky – measure length – shadow – any obj.
- Shadow – shortest – sun – highest pt. – sky
- Activity –
  - Find – small flat area – gnd. – receive sunlight – day – fix stick – 1 m – vertically
  - Start obs. – 11:00 AM – every min. – mark dot – gnd. – tip – stick’s shadow – mark until 1:10 PM
  - **Identify** – shadow – shortest – find out time – count dots – **record** time – repeat ex. – few days
  - Find duration – solar day – find diff. – time – consecutive days



- Find avg. duration – day – nearly equal – 24 hrs.
- Moon phases – another natural cycle – longer than day
- Moon – 29.5 days – cycle – all phases – this cycle – moon phases – basis – **month** – another unit – measure time
- Next larger unit – measure time – related – natural cycle – seasons

- Remember – learning earlier – earth – revolve – sun – 365 $\frac{1}{4}$ days – 1 revolution
- Earth – 1 cycle – seasons – this time – define – **solar year**

Lunar calendars

- Ancient times – people notice – 1 cycle – seasons – 12 cycles – moon phases – 12 lunar months
- This way – **lunar calendars** – created – shortest unit – day – lunar month – 29.5 days – lunar yr. – 12 lunar months
- Moon phases – easy, sound way – track time
- BUT – lunar calendar – seasons – not sync. – same lunar month – every year
- Reason – season repeat – 365 days – lunar yr. - 354 days

Solar calendars

- Imp. – know – arrival – seasons – agro. Purpose – this need – sync. seasons – creation – **solar calendars**
- Gregorian calendar – used today – solar calendar – months – adjusted – 365 days
- Reason – Gregorian calendars – some month – 30 days – others – 31 days – February – 28 days
- Earth – quarter day extra – 365 days – 1 revolution
- Extra hrs – add up – 1 day every 4 yrs. – to adjust – solar calendars – add extra day – every 4 yrs. – **leap year**
- Gregorian calendar – yr. – divisible by 4 – extra leap day added
- Leap yr. – Feb. – 29 days – calendar – sync. well – seasons
- **A Step Further**
    - Earth – not exactly – $365\frac{1}{4}$ day – slightly less time – 1 spring equinox to next spring equinox
    - Add – 1 day – every 4 yrs. – help sync. – seasons – BUT – adds over time
    - Fix this – leap year – skipped every 100 yr.
    - BUT – skipping every 100 yr. – calendar – lag behind
    - Every 400 yr. – leap yr. – added back – keep calendar – match – seasons – longer periods
- **A Step Further**
    - Learnt earlier – seasons – earth revolution – sun – movement – spring equinox to winter equinox and back
    - Time – successive equinox – tropical yr. – Gregorian calendar – based on tropical yr.
    - Also learnt – earlier – stars – rise – sunset – change through yr. – reason – earth revolution
    - Time duration – same stars – rise again – sunset – sidereal yr. – also used – define – solar calendar
    - Sidereal yr. – longer than tropical yr. – 20 min. – takes longer – diff. – noticeable
    - Modern times – astronomers – sidereal yr. – keep track – earth pos. – orbit
- **Interesting fact** –
    - 1000s of yrs. – people – Indians – obs. sky – develop calendars
    - People – ancient times – no idea – earth revolve – no modern instruments – BUT – yrs. – careful obs. – notice patterns, cycles – natural events
    - People – determine – yr. length – approx. 365 days – allow them – create calendars
    - Ex. – careful obs. reveal – sun – not always – exact east
    - Summer – northward – winter – southward – extremes – solstices – June 21, December 21
    - Sun – northward movement – December to June – *Uttarayan* – southward movement – June to December – *Dakshinayan*
    - Cycle – repeat yearly – close links – changing seasons
    - *Taittiriya Samhita* – records – verse 6.5.3 –

        - तस्‍ मादादि त्‍ य: षण्मासा दक्षिणेनैति षडुत्तरेण
            - Thus – sun – move southward – 6 months – move northwards – 6 months
    - Past – equinoxes, solstices – tracked – identify stars – rise – sunset
    - Ancient Indian texts – *Surya Siddhanta* – note – star pattern Capricorn (*Makar*) – bg. – sun – winter solstice – ancient times
        - नोर्मकरसंस् ङ्क्रान्तेः षण्मासा उत्तरायणम्। कर्क्कादेस्तु तथैव स्यात् षण्मासा दक्षिणायनम्।।9।।
            - Translation – moment – sun enters – constellation Capricorn – 6 months – northward progress (*Uttarayan*)
            - Likewise – moment – sun enters – constellation Cancer – 6 months – southward progress – (*Dakshinayan*)
    - Over yrs. – diff. calendars – evolved – specific needs – many of these – used – diff. parts – India – track time, celebrate festivals

Luni-solar calendars

- Another kind – calendar – use – moon phases – count days, months
- BUT also – make adjustments – stay in sync. – season cycles
- 12 lunar months – add up – 354 days – fall short – 11 days – solar yr. – every 2-3 yrs. – accumulated diff. – close – full month
- Every few yrs. – extra month (*Adhika Masa*) – intercalary month – add – yr. – some calendars
- Solar yr., lunar cycle – in step – such calendars – **luni-solar calendars**
- Combine elements – solar, lunar calendars – used – many parts – India
- **Interesting fact** –
    - Heard names – months – various Indian luni-solar calendars – *Chaitra*, *Vaishaka*, *Jyeshtha*, *Ashadha*, etc.
    - Some comm. – new month – start – 1st day after new moon – ends – new moon
    - Such calendars – *Amant*
    - Other comm. – start – new month – 1st day after full moon – ends – full moon
    - Such calendars – *Purnimant*

The Indian National Calendar

- National calendar – govt. of India – used along – Gregorian calendar – official purposes
- Solar calendar – 365 days – 1 yr. – begins – 22 march – day after spring equinox
- Unlike Gregorian – months – Indian calendar – 30 / 31 days
- Names – these months – taken – traditional Indian calendars
- Regular yr. – second to sixth month – 31 days – rest – 30 days
- Leap yr. – matched – Gregorian calendar – add day – Chaitra – 1st month – yr.
- Such yrs. – new yr. – 21 march – Gregorian calendar
- **Interesting fact** –
    - 1952 – govt. of India – set up – Calendar Reform Committee (CRC) – examine – existing calendars – recommend – accurate, uniform calendar – entire India
    - CRC – recommend – Unified National Calendar – adopted – use – 21 march 1956 CE – 1 Chaitra1878 Saka
    - Indian National Calendar – follow – gen. principles – *Surya Siddhanta*

Are Festivals Related to Astronomical Phenomenon?

- Indian festivals – tied – moon phases – hence – depend – lunar OR luni-solar calendars
- Ex. –
    - *Diwali* – new moon – *Kartika*
    - *Holi* – full moon – *Phalguna*
    - *Buddha Purnima* – full moon – *Vaishakha*
    - *Eid-ul-fitr* – sighting – crescent moon – end – *Ramazan*
    - *Dussehra* – 10th day – *Ashwina*
- This reason – festivals – diff. dates – Gregorian calendar – successive yr.
- Festivals – based – luni-solar calendars – Gregorian dates – shift – less than a month
- Luni-solar calendar – add month – few yrs. – correct diff. – lunar, solar yr.
- Contrast – purely lunar calendar – no repair – diff.
- Any festival – based – moon phases – *Eid-ul-fitr* – occur – diff. months – Gregorian calendar – successive yr.
- **A Step Further**
    - Few festivals – India – *Makar Sankranti*, *Pongal*, *Bihu*, *Vaisakhi*, *Poila Baisakha*, *Puthandu* – follow – solar sidereal calendar
    - These festivals – almost same date – every yr. – Gregorian calendar – based – tropical yr.
    - Long time ago – festivals – tied – solstice, equinox
    - Small diff. – sidereal, tropical yrs. – dates – festivals – shift away – solstices, equinoxes
    - Reason – slow wobble – Earth's axis – similar – wobbling top
    - Causes – dates – festivals – based – sidereal calendar – move ahead – tropical calendar
    - Ex. – *Makar Sankranti* – move ahead – 1 day – every 71 yrs.
- **Interesting fact** –
    - Dates – Indian Festivals – based – exact Lunar phase – sunrise
    - Sunrise – earlier – Eastern India – later – Western India – dates – shift – 1 day – within regions – same yr.
    - Maintain uniformity – Positional Astronomy Centre – Indian govt. – publish – Rashtriya panchang – detailed calculation – pos. – central obj. – moon, sun – central loc. – India
    - Based – these calc. – provide – adv. intimation – dates – festivals – Indian govt. – holiday declaration
- **Interesting fact** –
    - Moon, moonlight – inspire raga – Indian classical music
    - *Chandrakauns*, *Chandranandan*, *Shubhapantuvarali* – few ragas – display moon's imagery – their names – melodic expressions
    - Similar – mudras – ex. – *Chandrakala*, *Ardhachandran* – relate – moon – found – Indian classical dance – Bharatanatyam
    - Same – true – dance forms – Kathak, Odissi, Kuchipudi
    - Traditional painting styles – Madhubani, Warli, other forms – sculpture, pottery – Saura, Gond, other tribes – invoke depiction – moon, sun – imply – significance – daily life

Why Do We Launch Artificial Satellites in Space?

- Moon – natural satellite – orbit planet
- Beside moon – man-made satellites – various satellites – orbit earth
- **Artificial satellites** – visible – tiny specks – moving – night sky
- Most orbit – 800 km above earth's surface – 100 minutes – 1 orbit

- Satellites – help us – many ways – comm., navigation, weather monitoring, disaster mgmt., scientific research
- Indian Space Research Organisation (ISRO) – launch many satellites – support act.
- **Interesting fact** –
    - CartoSat series – satellites – ISRO – high-quality img. – earth – improve maps, plan cities, handle natural disasters
    - Mapping platform – Bhuvan – use img. – show terrain, soil, land use, vegetation, etc.
    - AstroSat – ISRO mission – scientific obs. – stars, celestial obj.
    - Another space mission – Chandrayaan1, 2, 3 – moon – Aditya L1 – sun – Mangalyaan – Mars
    - ISRO – lets – Indian students – build, launch satellites – AzaadiSat, InspireSat-1, Jugnu
- Activity –
    - Spotting – artificial satellite – night sky – before sunrise, sunset – go to loc. – clear view – sky – no obstruction – trees, tall buildings – accompanied by adult
    - Identify satellites – sky – look – moving obj. – sky – visible – light pt. – steady, flickering brightness – fast moving – visible – naked eye, binoculars
    - Use mobile apps, websites – provide details – satellites – visible – your loc. – pass above – sky
- **A Step Further**
    - Artificial satellites – sent up – space – many countries
    - Useful life – over – many satellites, rockets – space junk, debris
    - Debris – crowd space – can collide – working satellite
    - Small debris – burn up – atm. – fall toward earth – larger pieces – crash – gnd.
    - Countries – work together – remove – dangerous debris